equations simultaneously. A system of equations is called homogeneous if all the constant terms of R. H. S. Vanish

i. e. $b_1 = b_2 = \cdots = b_m = 0$

Otherwise the system is called non-homogeneous system.

There are number of numerical methods for solving a system of equations. These numerical methods are of two types namely :

(i) direct (ii) iterative

We will study a few methods below :

5.A.2. Gauss–Elimination Method (Direct Method) :

[NUH–2004, 2006, 2009, 2012]

This is a direct method based on the elimination of the unknowns by combining equations such that n equations in n unknowns are reduced to an equivalent upper triangular system which can be solved by back substitution.

Consider a system of n linear equations with n unknowns as given below :

[ইহা একটি সরাসরি প্রতিস্থাপন পদ্ধতি যা n সংখ্যক চলক বিশিষ্ট n সংখ্যক সমীকরণকে তার সমতুল্য upper triangular সমীকরণ জোটে রূপান্তর করে back substitution এর মাধ্যমে সমাধান করা হয়।

n সংখ্যক চলক বিশিষ্ট একটি n সংখ্যক সমীকরণের জোট বিবেচনা করি।]

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \cdots \quad \cdots \quad \cdots \quad \cdots &\\ \cdots \quad \cdots \quad \cdots \quad \cdots &\\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}\right\} \cdots\cdots (1)$$

The system (1) is equivalent to [জোট (1) এর সমতুল্য]

$$AX = B \cdots\cdots (2)$$

where [যেখানে] $A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$, $X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ and $B = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$

Now our aim is to reduce the augmented matrix of (1) to upper triangular matrix [আমাদের উদ্দেশ্য হল বর্ধিত matrix কে upper triangular matrix এ রূপান্তর করা।]

Now, the augmented matrix (A|B) is [এখন বর্ধিত ম্যাট্রিক্স (A|B) হল]

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array}\right) \quad \cdots\cdots (3)$$

Now dividing the first row by a_{11} (if $a_{11} = 0$, the rows of the matrix are rearranged in a suitable way) and then subtract this row multiplied by $a_{21}, a_{31}, \cdots, a_{n1}$ from the second, third, $\cdots$, nth rows respectively. Then (3) becomes [এখন ১ম সারিকে $a_{11}(a_{11} \neq 0)$ দ্বারা ভাগ করি। (যদি $a_{11} = 0$ তবে সমীকরণ গুলিকে পুনঃসজ্জিত করি যাতে x_1, এর সহগ শূন্য না হয়। এই পদ্ধতিকে Partial pivoting বলে।) তার পরে ১ম সমীকরণকে $a_{21}, a_{31}, \cdots, a_n$, দ্বারা গুণ করে যথাক্রমে ২য়, ৩য় ... n তম সারি হতে বিয়োগ করে। (3) সমীকরণ হবে।]

$$\left(\begin{array}{cccc|c} 1 & a'_{12} & \cdots & a'_{1n} & b'_1 \\ 0 & a'_{22} & \cdots & a'_{2n} & b'_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & a'_{n2} & \cdots & a'_{nn} & b'_n \end{array}\right) \quad \cdots\cdots (4)$$

Now dividing 2nd row of (4) by a'_{22} (if $a'_{22} = 0$, the rows are arranged in a suitable way). and then we subtract this row multiplied by a'_{32} , $\cdots a'_{n2}$ from the third, fourth, $\cdots$, nth rows. We continue this procedure as far as possible, all elements below the leading diagonal elements of A are made to zero and finally we obtain the following reduced matrix [এখন (4) নং এ ২য় সারিকে a'_{22} দ্বারা ভাগ করি। (যদি $a'_{22} = 0$ হয় তবে পরবর্তী সারিগুলির সাথে পুনঃসজ্জিত করি। এই পদ্ধতিকে Partial Pivoting বলে।) তার পরে ২য় সারিকে $a'_{32}, a'_{42}, \cdots, a'_{n2}$ দ্বারা পর্যায় ক্রমে গুণ করি এবং তা ৪র্থ , ..., nতম সারি হতে বিয়োগ করি।) এই পদ্ধতীতে অগ্রসর হই যেন প্রধান কর্নের নিচের সকল উপাদান শূন্য হয়। সর্বশেষ আমরা নিম্নরূপ ম্যাট্রিক্স পাব।]

$$\left(\begin{array}{ccccc|c} 1 & a'_{12} & a'_{13} & \cdots & a'_{1n} & b'_1 \\ 0 & 1 & a''_{23} & \cdots & a''_{2n} & c'_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 1 & k'_n \end{array}\right)$$

∴ The reduced system is [রূপান্তরিত জোট]

$$\left.\begin{aligned} x_1 + a'_{12}x_2 + a'_{13}x_3 + \cdots + a'_{1n}x_n &= b'_1 \\ x_2 + a''_{23}x_3 + \cdots + a''_{2n}x_n &= c'_2 \\ \cdots \quad \cdots \quad \cdots \quad \cdots & \\ x_n &= k'_n \end{aligned}\right\} \quad \cdots\cdots (5)$$

Now, by back substitution, we solve for $x_n, x_{n-1}, \cdots x_2, x_1$ [এ back substitution পদ্ধতিতে সমাধান করে $x_n, x_{n-1}, \cdots, x_2, x_1$ এর মান নি করি।]

5.A.3. Gauss–Jordan (Gauss-Jacobi) Method :

[NUH–2000, 2010, 2011, 2012, BSc(Pass)–2007, 2009, 201

This method is a modification of the Gauss eliminati method. In this method, the coefficient matrix A of the syste AX = B is brought to a unit matrix.

Consider a system of n linear equations with n unknowns given below :

[ইহা গাউস-প্রতিস্থাপন পদ্ধতির সংশোধিত রূপ। এই পদ্ধতিতে সমীকরণ জোটের স ম্যাট্রিক্স A কে একটি একক (unit) ম্যাট্রিক্সে রূপান্তরিত করা হয়।

n সংখ্যক চলক বিশিষ্ট একটি n সংখ্যক সমীকরণের জোট বিবেচনা করি।]

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \cdots \quad \cdots \quad \cdots \quad \cdots & \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}\right\} \quad \cdots\cdots (1)$$

The system (1) is equivalent to [সমীকরণ (1) এর সমতুল্য জোট]

$$AX = B \cdots\cdots (2)$$

where [যেখানে] $A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}, X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} B = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$

Now the augmented matrix of the above system is [উপ সমীকরণ জোটের বর্ধিত ম্যাট্রিক্স হল]

$$(A|B) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array}\right) \cdots\cdots (3)$$

$u_{11} = a_{11}$ $\quad u_{12} = a_{12}$ $\quad u_{13} = a_{13}$

$l_{21}\, u_{11} = a_{21}$ $\quad l_{21}\, u_{12} + u_{22} = a_{22}$ $\quad l_{21}\, u_{13} + u_{23} = a_{23}$

$l_{31}\, u_{11} = a_{31}$ $\quad l_{31}\, u_{12} + l_{32}\, u_{22} = a_{32}$ $\quad l_{31}\, u_{13} + l_{32}\, u_{23} + u_{33} = a_{33}$

$u_{11} = a_{11}$ $\quad u_{12} = a_{12}$ $\quad u_{13} = a_{13}$

$\Rightarrow l_{21} = \dfrac{a_{21}}{a_{11}}$ $\quad u_{22} = a_{22} - \dfrac{a_{21}}{a_{11}} \cdot a_{12}$ $\quad u_{23} = a_{23} - \dfrac{a_{21}}{a_{11}} \cdot a_{13}$

$l_{31} = \dfrac{a_{31}}{a_{11}}$ $\quad l_{32} = \dfrac{a_{32} - \dfrac{a_{31}}{a_{11}} \cdot a_{12}}{a_{22} - \dfrac{a_{21}}{a_{11}} \cdot a_{12}}$

$$u_{33} = a_{33} - \frac{a_{31}}{a_{11}} \cdot a_{13} - \left(\frac{a_{32} - \dfrac{a_{31}}{a_{11}} \cdot a_{12}}{a_{22} - \dfrac{a_{21}}{a_{11}} \cdot a_{12}} \right) \left(a_{23} - \frac{a_{21}}{a_{11}} \cdot a_{13} \right)$$

therefore L and U are known.

WORKED OUT EXAMPLES

Example–1. Solve by Gauss-Elemination method :

$$x - y + z = 1, -3x + 2y - 3z = -6, 2x - 5y + 4z = 5.$$

Solution : Given the system of equations

$$\left.\begin{aligned} x - y + z &= 1 \\ -3x + 2y - 3z &= -6 \\ 2x - 5y + 4z &= 5 \end{aligned}\right\} \ \ldots\ldots (1)$$

The system (1) is equivalent to

$$AX = B \ \ldots\ldots (2)$$

where $A = \begin{bmatrix} 1 & -1 & 1 \\ -3 & 2 & -3 \\ 2 & -5 & 4 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ -6 \\ 5 \end{bmatrix}$

Now our aim is to reduce the augmented matrix of (1) to upper triangular matrix. Now the augmented matrix (A | B) is :

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ -3 & 2 & -3 & -6 \\ 2 & -5 & 4 & 5 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & -1 & 0 & -3 \\ 0 & -3 & 2 & 3 \end{array}\right), \quad \begin{array}{l} r_2' = r_2 + 3r_1 \\ r_3' = r_3 - 2r_1 \end{array}$$

$$r_2' = -r_2 \sim \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & -3 & 2 & 3 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 2 & 12 \end{array}\right), \quad r_3' = r_3 + 3r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 6 \end{array}\right), \quad r_3' = \frac{1}{2} r_3$$

$\therefore$ The reduced system is

$$x - y + z = 1$$
$$y + 0z = 3$$
$$z = 6$$

Now, by back substitution, we get

$$z = 6$$
$$y = 3$$
$$x - 3 + 6 = 1$$
$$\Rightarrow x = 1 - 3 = -2$$

Hence, the required solution is $x = -2$, $y = 3$ and $z = 6$.

Example–2. Solve by Gauss-Elimination Method :

$x + 3y + 10z = 24$, $2x + 17y + 4z = 35$, $28x + 4y - z = 32$.

Solution : Given the system of equations

$$\left.\begin{array}{l} x + 3y + 10z = 24 \\ 2x + 17y + 4z = 35 \\ 28x + 4y - z = 32 \end{array}\right\} \ldots\ldots (1)$$

The system (1) is equivalent to

$$AX = B \ldots\ldots (2)$$

where $A = \begin{bmatrix} 1 & 3 & 10 \\ 2 & 17 & 4 \\ 28 & 4 & -1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B \begin{bmatrix} 24 \\ 35 \\ 32 \end{bmatrix}$

Now our aim is reduce the augmented matrix of (1) to triangular matrix :

Now the augmented matrix (A | B) is :

$$\left(\begin{array}{ccc|c} 1 & 3 & 10 & 24 \\ 2 & 17 & 4 & 35 \\ 28 & 4 & -1 & 32 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 3 & 10 & 24 \\ 0 & 11 & -16 & -13 \\ 0 & -80 & -281 & -640 \end{array}\right), \quad \begin{array}{l} r_2' = r_2 - 2r_1 \\ r_3' = r_3 - 28r_1 \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 3 & 10 & 24 \\ 0 & 1 & -16/11 & -13/11 \\ 0 & -80 & -281 & -640 \end{array}\right), \quad r_2' = \frac{1}{11} r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 3 & 10 & 24 \\ 0 & 1 & -16/11 & -13/11 \\ 0 & 0 & -4371/11 & -8080/11 \end{array}\right), \quad r_3' = r_3 + 80r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 3 & 10 & 24 \\ 0 & 1 & -16/11 & -13/11 \\ 0 & 0 & 1 & 8080/4371 \end{array}\right), \quad r_3' = -\frac{11}{4371} r_2$$

$\therefore$ The reduced system is

$$x + 3y + 10z = 24$$

$$y - \frac{16}{11} z = -\frac{13}{11}$$

$$z = \frac{8080}{4371}$$

Now, by back substitution, we get

$$z = \frac{8080}{4371}$$

$$y = -\frac{13}{11} + \frac{16}{11} \times \frac{8080}{11371} = \frac{6587}{4371}$$

$$x = 24 - 3 \times \frac{6587}{4371} - 10 \times \frac{8080}{4371} = \frac{4343}{4371}$$

Hence, the required solution is

$$x = \frac{4343}{4371}, \quad y = \frac{6587}{4371} \text{ and } z = \frac{8080}{4371}.$$

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Example–3. Solve by Gauss-Elimination method :

$10x + y + z = 12,\ x + 10y + z = 12,\ x + y + 10z = 12.$

Solution : Given the system of equations

$$\left.\begin{aligned} 10x + y + z &= 12 \\ x + 10y + z &= 12 \\ x + y + 10z &= 12 \end{aligned}\right\} \ldots\ldots (1)$$

The system (1) is equivalent to

$$AX = B \ldots\ldots (2)$$

where $A = \begin{bmatrix} 10 & 1 & 1 \\ 1 & 10 & 1 \\ 1 & 1 & 10 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B \begin{bmatrix} 12 \\ 12 \\ 12 \end{bmatrix}$

Now, our aim is to reduce the augmented matrix of (1) to upper triangular matrix :

Now, the augmented matrix (A|B) is :

$$\left(\begin{array}{ccc|c} 10 & 1 & 1 & 12 \\ 1 & 10 & 1 & 12 \\ 1 & 1 & 10 & 12 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1/10 & 1/10 & 12/10 \\ 1 & 10 & 1 & 12 \\ 1 & 1 & 10 & 12 \end{array}\right), \quad r_1' = \frac{1}{10} r_1$$

$$\sim \left(\begin{array}{ccc|c} 1 & \frac{1}{10} & \frac{1}{10} & \frac{12}{10} \\ 0 & \frac{99}{10} & \frac{9}{10} & \frac{108}{10} \\ 0 & \frac{9}{10} & \frac{99}{10} & \frac{108}{10} \end{array}\right), \quad \begin{aligned} r_2' &= r_2 - r_1 \\ r_3' &= r_3 - r_1 \end{aligned}$$

$$\sim \left(\begin{array}{ccc|c} 1 & \frac{1}{10} & \frac{1}{10} & \frac{12}{10} \\ 0 & 1 & \frac{1}{11} & \frac{108}{99} \\ 0 & \frac{9}{10} & \frac{99}{10} & \frac{108}{10} \end{array}\right), \quad r_2' = \frac{10}{99} r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & \frac{1}{10} & \frac{1}{10} & \frac{12}{10} \\ 0 & 1 & \frac{1}{11} & \frac{108}{99} \\ 0 & 0 & \frac{1080}{110} & \frac{9720}{990} \end{array}\right), \quad r_3' = r_3 - \frac{9}{10} r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & \frac{1}{10} & \frac{1}{10} & \frac{12}{10} \\ 0 & 1 & \frac{1}{11} & \frac{108}{99} \\ 0 & 0 & 1 & 1 \end{array}\right), \quad r_3' = \frac{110}{1080} r_3$$

$\therefore$ The required system is

$$x + \frac{1}{10}y + \frac{1}{10}z = \frac{12}{10}$$

$$y + \frac{1}{11}z = \frac{108}{99}$$

$$z = 1$$

Now, by back substitution, we get

$$z = 1$$

$$y = \frac{108}{99} - \frac{1}{11} \cdot 1 = 1$$

$$x = \frac{12}{10} - \frac{1}{10} \cdot 1 - \frac{1}{10} \cdot 1 = \frac{6}{5} - \frac{1}{5} = 1$$

$\therefore$ The required solution is

$$x = 1, \ y = 1 \text{ and } z = 1.$$

Example–4. Solve by Gauss-Elimination method :

$$x - 3y - z = -30, \ 2x - y - 3z = 5, \ 5x - y - 2z = 142.$$

Solution : Given the system of equations

$$\left.\begin{array}{l} x - 3y - z = -30 \\ 2x - y - 3z = 5 \\ 5x - y - 2z = 142 \end{array}\right\} \ldots\ldots (1)$$

The system (1) is equivalent to

$$AX = B \ldots\ldots (2)$$

where $A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & -1 & -3 \\ 5 & -1 & -2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} -30 \\ 5 \\ 142 \end{bmatrix}$

Now, our aim is to reduce the augmented matrix of (1) to upper angular matrix.

Now the augmented matrix (A | B) is :

$$\left(\begin{array}{ccc|c} 1 & -3 & -1 & -30 \\ 2 & -1 & -3 & 5 \\ 5 & -1 & -2 & 142 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & -3 & -1 & -30 \\ 0 & 5 & -1 & 65 \\ 0 & 14 & 3 & 292 \end{array}\right), \begin{array}{l} r_2' = r_2 - 2r_1 \\ r_3' = r_3 - 5r_1 \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -3 & -1 & -30 \\ 0 & 1 & -1/5 & 13 \\ 0 & 14 & 3 & 292 \end{array}\right), \quad r_2' = \frac{1}{5} r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & -3 & -1 & -30 \\ 0 & 1 & -1/5 & 13 \\ 0 & 0 & 29/5 & 110 \end{array}\right), \quad r_3' = r_3 - 14r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & -3 & -1 & -30 \\ 0 & 1 & -1/5 & 13 \\ 0 & 0 & 1 & 550/29 \end{array}\right), \quad r_3' = \frac{5}{29} r_3$$

∴ The required system is

$$x - 3y - z = -30$$

$$y - \frac{1}{5} z = 13$$

$$z = \frac{550}{29}$$

Now, by back substitution, we get

$$z = \frac{550}{29}$$

$$y = 13 + \frac{110}{29} = \frac{487}{29}$$

$$x = -30 + 3.\frac{87}{29} + \frac{550}{29} = \frac{1141}{29}$$

Hence, the required solution is

$$x = \frac{1141}{29}, \quad y = \frac{487}{29} \text{ and } z = \frac{550}{29}.$$

Example–5. Solve by Gauss-Elimination method :

$3x + y - z = 3,\ 2x - 8y + z = -5,\ x - 2y + 9z = 8.$

Solution : Given the system of equations

$$\left.\begin{aligned} 3x + y - z &= 3 \\ 2x - 8y + z &= -5 \\ x - 2y + 9z &= 8 \end{aligned}\right\} \quad \ldots\ldots (1)$$ **[NUH–14, BSc(Pass)–10]**

The system (1) is equivalent to

$$AX = B \quad \ldots\ldots (2)$$

where, $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -8 & 1 \\ 1 & -2 & 9 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 3 \\ -5 \\ 8 \end{bmatrix}$

Now, our aim is to reduce the augmented matrix of (1) to upper triangular matrix.

Now the augmented matrix $(A|B)$ is :

$$\left(\begin{array}{ccc|c} 3 & 1 & -1 & 3 \\ 2 & -8 & 1 & -5 \\ 1 & -2 & 9 & 8 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1/3 & -1/3 & 1 \\ 2 & -8 & 1 & -5 \\ 1 & -2 & 9 & 8 \end{array}\right), \quad r_1' = \frac{1}{3} r_1$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1/3 & -1/3 & 1 \\ 0 & -26/3 & 5/3 & -7 \\ 0 & -7/3 & 29/3 & 7 \end{array}\right), \quad \begin{aligned} r_2' &= r_2 - 2r_1 \\ r_3' &= r_3 - r_1 \end{aligned}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1/3 & -1/3 & 1 \\ 0 & 1 & -5/26 & 21/26 \\ 0 & -7/3 & 28/3 & 7 \end{array}\right), \quad r_2' = -\frac{3}{26} r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1/3 & -1/3 & 1 \\ 0 & 1 & -5/26 & 21/26 \\ 0 & 0 & 693/78 & 693/78 \end{array}\right), \quad r_3' = r_3 + \frac{7}{3} r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1/3 & -1/3 & 1 \\ 0 & 1 & -5/26 & 21/26 \\ 0 & 0 & 1 & 1 \end{array}\right), \quad r_3' = \frac{78}{693} r_3$$

$\therefore$ The required system is

$$x + \frac{1}{3} y - \frac{1}{3} z = 1$$

$$y - \frac{5}{26} z = \frac{21}{26}$$

$$z = 1$$

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Now, by back substitution, we get

$$z = 1$$

$$y = \frac{21}{26} + \frac{5}{26} \cdot 1 = 1$$

$$x = 1 - \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 1 = 1$$

Hence, the required solution is

$$x = 1, y = 1 \text{ and } z = 1.$$

Example–6. Solve by Gauss -Elemination method :

$$5x + 3y - 3z = -1,\ 3x + 2y - 2z = -1,\ 2x - y + 2z = 8.$$

Solution : Given the system of equation

$$\left.\begin{aligned} 5x + 3y - 3z &= -1 \\ 3x + 2y - 2z &= -1 \\ 2x - y + 2z &= 8 \end{aligned}\right\} \ldots\ldots (1)$$

The system (1) is equivalent to

$$AX = B \ldots\ldots (2)$$

where, $A = \begin{bmatrix} 5 & 3 & -3 \\ 3 & 2 & -2 \\ 2 & -1 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B \begin{bmatrix} -1 \\ -1 \\ 8 \end{bmatrix}$

Now our aim is to reduce the augmented matrix of (1) to upper ngular matrix.

Now, the augmented matrix (A|B) is :

$$\left(\begin{array}{ccc|c} 5 & 3 & -3 & -1 \\ 3 & 2 & -2 & -1 \\ 2 & -1 & 2 & 8 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 3/5 & -3/5 & -1/5 \\ 3 & 2 & -2 & -1 \\ 2 & -1 & 2 & 8 \end{array}\right), \quad r_1' = \frac{1}{5} r_1$$

$$\sim \left(\begin{array}{ccc|c} 1 & 3/5 & -3/5 & -1/5 \\ 0 & 1/5 & -1/5 & -2/5 \\ 0 & -11/5 & 16/5 & 42/5 \end{array}\right), \quad \begin{aligned} r_2' &= r_2 - 3r_1 \\ r_3' &= r_3 - 2r_1 \end{aligned}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 3/5 & -3/5 & -1/5 \\ 0 & 1 & -1 & -2 \\ 0 & -11/5 & 16/5 & 42/5 \end{array}\right), \quad r_2' = 3r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 3/5 & -3/5 & -1/5 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 4 \end{array}\right), \quad r_3' = r_3 + \frac{11}{5} r_2$$

$\therefore$ The required system is

$$x+\frac{3}{5}y-\frac{3}{5}z=-\frac{1}{5}$$
$$y-z=-2$$
$$z=4$$

Now, by back substitution, we get

$$z=4$$
$$y=-2+4=2$$
$$x=-\frac{1}{5}-\frac{3}{5}.2+\frac{3}{5}.4=1$$

Hence, the required solution is $x = 1, y = 2$ and $z = 4$.

Example–7. Solve by Gauss-Elimination method :

$$2x - 3y + z = -1,\ x + 4y + 5z = 25,\ 3x - 4y + z = 2.$$

[NUH–2013(Sam

Solution : Given the system of equations

$$\left.\begin{aligned} 2x - 3y + z &= -1 \\ x + 4y + 5z &= 25 \\ 3x - 4y + z &= 2 \end{aligned}\right\} \ldots\ldots (1)$$

The system (1) is equivalent to

where, $A = \begin{bmatrix} 2 & -3 & 1 \\ 1 & 4 & 5 \\ 3 & -4 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B \begin{bmatrix} -1 \\ 25 \\ 2 \end{bmatrix}$

Now, our aim is to reduce the augmented matrix of (1) to u triangular matrix.

Now, the augmented matrix $(A|B)$ is.

$$\left(\begin{array}{ccc|c} 2 & -3 & 1 & -1 \\ 1 & 4 & 5 & 25 \\ 3 & -4 & 1 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -3/2 & 1/2 & -1/2 \\ 1 & 4 & 5 & 25 \\ 3 & -4 & 1 & 2 \end{array}\right), \quad r_1' = \frac{1}{2}r_1$$

$$\sim \left(\begin{array}{ccc|c} 1 & -\frac{3}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{11}{2} & \frac{9}{2} & \frac{51}{11} \\ 0 & \frac{1}{2} & -\frac{1}{2} & \frac{7}{2} \end{array}\right), \quad \begin{aligned} r_2' &= r_2 - r_1 \\ r_3' &= r_3 - 3r_1 \end{aligned}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -\frac{3}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & \frac{9}{11} & \frac{51}{11} \\ 0 & \frac{1}{2} & -\frac{1}{2} & \frac{7}{2} \end{array}\right), \quad r_2' = \frac{2}{11} r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & -\frac{3}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & \frac{9}{11} & \frac{51}{11} \\ 0 & 0 & -\frac{20}{22} & \frac{26}{22} \end{array}\right), \quad r_3' = r_3 - \frac{1}{2} r_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & -\frac{3}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & \frac{9}{11} & \frac{51}{11} \\ 0 & 0 & 1 & -\frac{26}{20} \end{array}\right), \quad r_3' = -\frac{22}{20} r_3$$

∴ The required system is

$$x - \frac{3}{2}y + \frac{1}{2}z = -\frac{1}{2}$$

$$y + \frac{9}{11}z = \frac{51}{11}$$

$$z = -\frac{26}{20}$$

Now, by back substitution, we get

$$z = -\frac{26}{20} = -\frac{13}{10}$$

$$y = \frac{51}{11} - \frac{9}{11} \cdot \left(-\frac{13}{10}\right) = \frac{57}{10}$$

$$x = -\frac{1}{2} + \frac{9}{2} \cdot \frac{57}{10} - \frac{1}{2}\left(-\frac{13}{10}\right) = \frac{87}{10}$$

Hence, the required solution is

$$x = \frac{87}{10}, y = \frac{57}{10} \text{ and } z = -\frac{13}{10}.$$

Example–8. Solve by Gauss-Elimination method :

$$x + 2y + 3z = 6,\ 2x + 4y + z = 7,\ 3x + 2y + 9z = 14.$$

Solution : Given the system of equations

$$\left.\begin{aligned} x + 2y + 3z &= 6 \\ 2x + 4y + z &= 7 \\ 3x + 2y + 9z &= 14 \end{aligned}\right\} \ldots\ldots (1)$$

The system (1) is equivalent to

$$AX = B \ldots\ldots (2)$$

where, $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 2 & 9 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 6 \\ 7 \\ 14 \end{bmatrix}$

Now, our aim is to reduce the augmented matrix of (1) to upper triangular matrix.

Now, the augmented matrix $(A|B)$ is :

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & 4 & 1 & 7 \\ 3 & 2 & 9 & 14 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 0 & -5 & -5 \\ 0 & -4 & 0 & -4 \end{array}\right), \begin{matrix} r_2' = r_2 - 2r_1 \\ r_3' = r_3 - 3r_1 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 0 & 5 & 5 \\ 0 & 4 & 0 & 4 \end{array}\right), \begin{matrix} r_2' = -r_2 \\ r_3' = -r_3 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 4 & 0 & 4 \\ 0 & 0 & 5 & 5 \end{array}\right), \quad r_2 \Leftrightarrow r_3$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right), \begin{matrix} r_2' = \frac{1}{4} r_2 \\ r_3' = \frac{1}{5} r_3 \end{matrix}$$

$\therefore$ The required system is

$$x + 2y + 3z = 6$$

$$y + 0z = 1$$

$$z = 1$$

Now, by back substitution, we get

$$z = 1$$

$$y = 1$$

$$x = 6 - 5 = 1$$

Hence, the required solution is $x = 1, y = 1$ and $z = 1$.

Example–9. Solve the following system of equations by Gauss elimination method :

$$x + \frac{1}{2}y + \frac{1}{3}z = 1$$

$$\frac{1}{2}x + \frac{1}{3}y + \frac{1}{4}z = 0$$

$$\frac{1}{3}x + \frac{1}{4}y + \frac{1}{5}z = 0$$

[NUH–2003]

Solution : Given that the system of equations

$$\left.\begin{aligned} x + \frac{1}{2}y + \frac{1}{3}z &= 1 \\ \frac{1}{2}x + \frac{1}{3}y + \frac{1}{4}z &= 0 \\ \frac{1}{3}x + \frac{1}{4}y + \frac{1}{5}z &= 0 \end{aligned}\right\} \quad \ldots\ldots (1)$$

The system (1) is equivalent to

$$AX = B \quad \ldots\ldots (2)$$

Where $A = \begin{pmatrix} 1 & 1/2 & 1/3 \\ 1/2 & 1/3 & 1/4 \\ 1/3 & 1/4 & 1/5 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

We shall solve the above system of equations by Gauss elimination method. So our aim is to reduce the augmented matrix of (1) to upper triangular matrix. Now the augmented matrix $(A|B)$ is

$$\left(\begin{array}{ccc|c} 1 & 1/2 & 1/3 & 1 \\ 1/2 & 1/3 & 1/4 & 0 \\ 1/3 & 1/4 & 1/5 & 0 \end{array}\right)$$

$r_2' = r_2 - \frac{1}{2}r_1$, $r_3' = r_3 - \frac{1}{3}r_1$

$$\left(\begin{array}{ccc|c} 1 & 1/2 & 1/3 & 1 \\ 0 & 1/12 & 1/12 & -1/2 \\ 0 & 1/12 & 4/45 & -1/3 \end{array}\right)$$

$$\xrightarrow{r_2' = 12 \times r_2} \left(\begin{array}{ccc|c} 1 & 1/2 & 1/3 & 1 \\ 0 & 1 & 1 & -6 \\ 0 & 1/12 & 4/45 & -1/3 \end{array}\right)$$

$$\xrightarrow{r_3' = r_3 - \frac{1}{12} r_2} \left(\begin{array}{ccc|c} 1 & 1/2 & 1/3 & 1 \\ 0 & 1 & 1 & -6 \\ 0 & 0 & 1/180 & 1/6 \end{array}\right)$$

$$\xrightarrow{r_3' = 180 \times r_3} \left(\begin{array}{ccc|c} 1 & 1/2 & 1/3 & 1 \\ 0 & 1 & 1 & -6 \\ 0 & 0 & 1 & 30 \end{array}\right)$$

$\therefore$ The reduced system is

$$\left.\begin{aligned} x + \frac{1}{2}y + \frac{1}{3}z &= 1 \\ y + z &= -6 \\ z &= 30 \end{aligned}\right\}$$

Now by back substitution, we get

$z = 30$

$y + 30 = -6 \Rightarrow y = -36$

$x + \frac{1}{2} \times (-36) + \frac{1}{3} \times 30 = 1$

or, $x - 18 + 10 = 1$

$\therefore x = 9.$

$\therefore$ The required solution is $x = 9, y = -36, z = 3.$

Example–10. Solve by Gauss-Jordan method :

$4x + y + 3z = 11$, $3x + 4y + 2z = 11$ and $2x + 3y + z = 7$. [BSc(Pass)-20

Solution : Given

$$\left.\begin{aligned} 4x + y + 3z &= 11 \\ 3x + 4y + 2z &= 11 \\ 2x + 3y + z &= 7 \end{aligned}\right\} \ldots\ldots (1)$$

The system (1) is equivalent to

$$AX = B \ldots\ldots (2)$$

where $A = \begin{pmatrix} 4 & 1 & 3 \\ 3 & 4 & 2 \\ 2 & 3 & 1 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $B = \begin{pmatrix} 11 \\ 11 \\ 7 \end{pmatrix}$